MedFactEval and MedAgenBrief: A Framework and Workflow for Generating and Evaluating Factual Clinical Summaries

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BACKGROUND

- Clinical documentation drives physician burnout; LLMs offer automation potential
- Factual errors (omissions and hallucinations) impede safe AI adoption in healthcare
- Expert review is too slow/expensive for continuous quality assurance
- Current metrics (BLEU, ROUGE) fail to capture clinical nuance
- Need: scalable evaluation framework replicating expert judgment

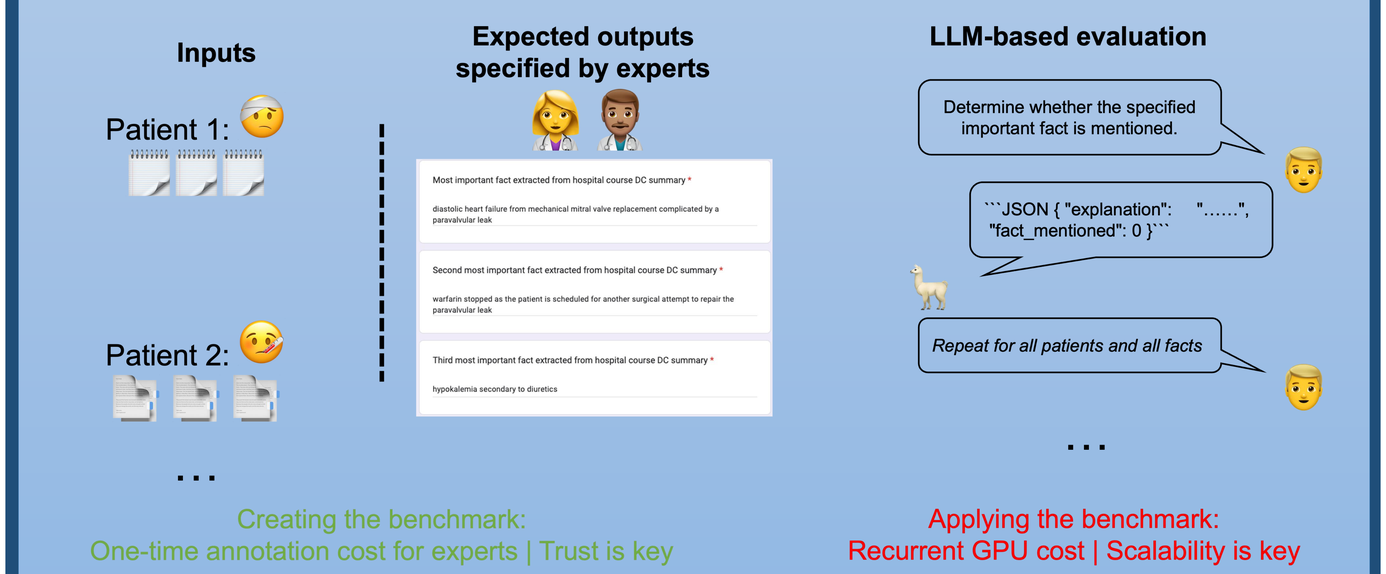
OBJECTIVE

- Introduce **MedFactEval**: fact-grounded evaluation framework using clinician-defined key facts
- Present **MedAgentBrief**: multi-step workflow for high-quality discharge summaries
- Validate LLM Jury agreement with seven-physician panel (Cohen's κ)
- Benchmark generation strategies on factual performance vs. cost

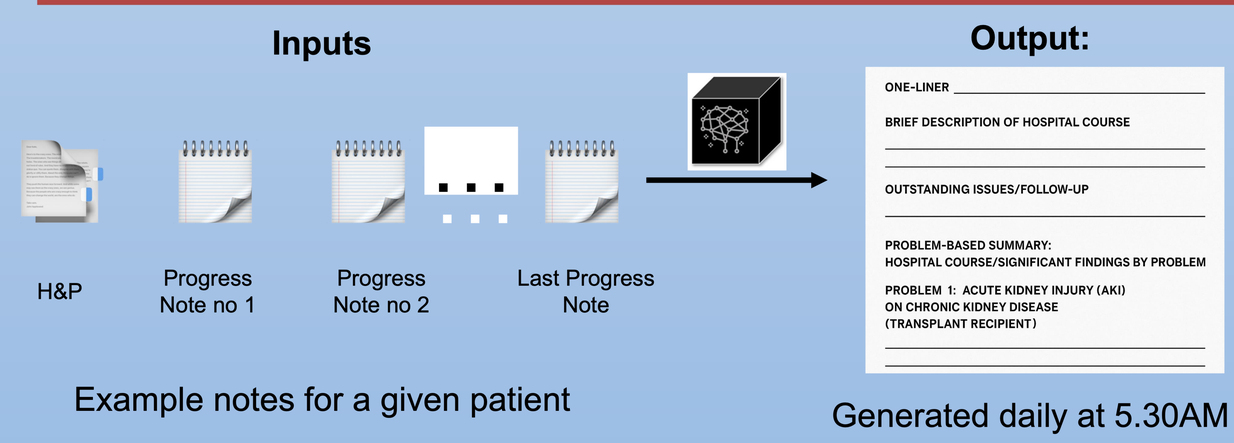
DATA

- **30 patient cases** – adult inpatients with 2–14 day stays at Stanford Health Care (Jan 2023–Jun 2024)
- **90 clinician-defined key facts** – 3 high-salience facts per case: diagnoses (51%), management changes (40%), follow-ups (6%)
- **12 progress notes per case (median, IQR 10–16)** – including physician and allied health documentation

MedFactEval: A Fact-Grounded, Scalable Approach to Evaluating AI-Generated Discharge Summaries



MedAgentBrief: From clinical notes to hospital course summaries



Example notes for a given patient

Generated daily at 5.30AM

Usage include:

- First draft discharge summary
- Handoffs
- Improved continuity of care

EXPERIMENTS

Systems compared

- Single-prompt baseline: one long prompt with all notes.
- MedAgentBrief: multi-step agentic workflow.

Meta-evaluation

- Seven attendings rated each summary with respect to key facts.
- Gold standard: majority vote of the 7 physicians.
- Compare:
  - Single physician vs. gold standard.

RESULTS

MedAgentBrief improves factuality: captures more clinician-defined key facts than a single-prompt baseline (e.g., +17 percentage points for GPT-4o).

LLM Jury ≈ expert: multi-model LLM jury matches or exceeds a single physician's agreement with the 7-physician gold standard (non-inferior on Cohen's κ).

Scalable evaluation: automated fact-based judging is ~100× faster and ~10,000× cheaper than manual review; main remaining failures are omissions, with contradictions in ~10–25% of cases.

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